

Exploratory Data Analysis on the Diabetes Dataset

Report

Introduction

The diabetes dataset consists of 11 variables. The full list includes age, sex, bmi, bp, s1-s6, and Progression. Age and sex are standardised.

The bp variable corresponds to the patients' blood pressure and bmi to the body mass index. Progression refers to the disease's progression one year after baseline (the target variable). The s1-s6 variables are standardised blood serum measurements, specifically:

- s1: Total serum cholesterol (TC)
- s2: Low-density lipoproteins (LDL)
- s3: High-density lipoproteins (HDL)
- s4: Total cholesterol/HDL ratio (TCH)
- s5: Log of serum triglycerides (LTG)
- s6: Blood glucose level (GLU)

The feature values are between about -0.15 and 0.2 - the predictors have already been standardised. This means, for example, a negative value for s1 is not 'negative cholesterol', rather it is cholesterol below the dataset average.

The dataset contains 11 columns and 442 rows (including the header row).

Data Cleaning

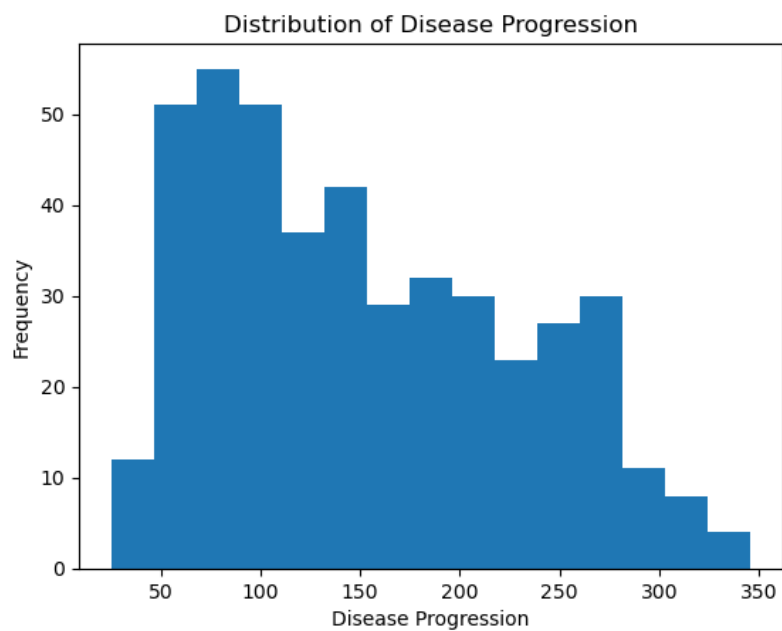
The dataset was inspected before analysis by checking the data types of each variable, searching for duplicate records and verifying that there were no impossible or unrealistic values. The dataset contained no duplicate rows, all variables were stored in appropriate numerical formats, and no impossible or unrealistic values were identified. Therefore, no further cleaning was required.

Missing Data

The dataset was searched for missing data using `df.isnull().sum()`. The output showed that there were 0 instances of missing data in each column. Therefore, there was no handling process.

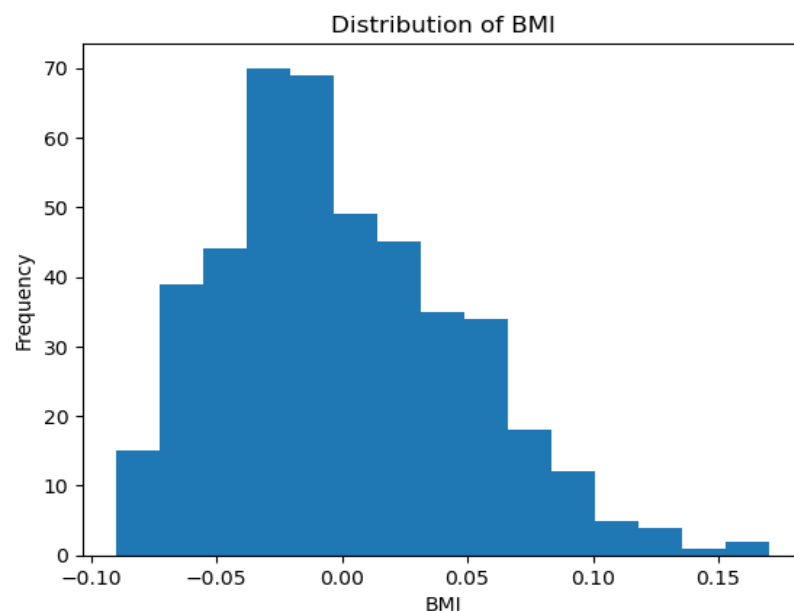
Data stories and Visualisations

Histogram of Progression



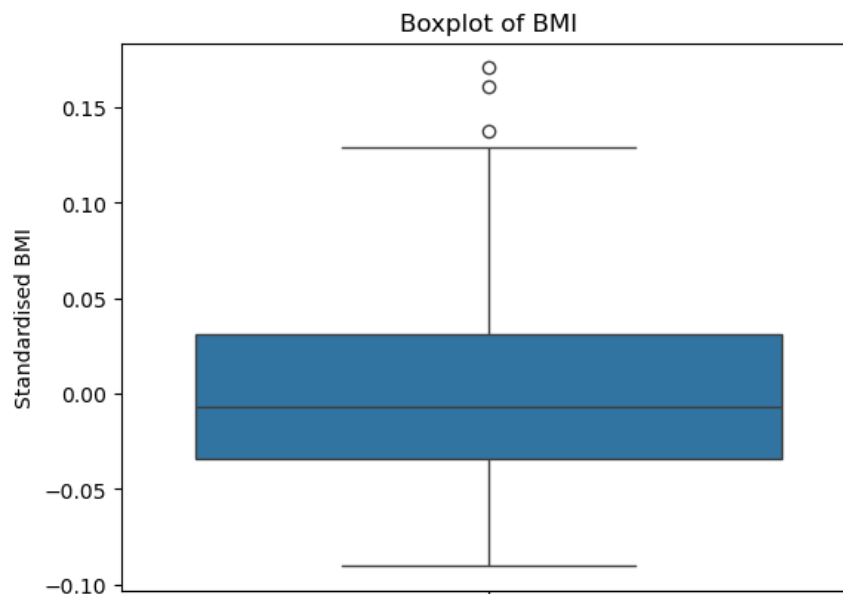
The histogram shows that disease progression is slightly positively skewed. Most patients have progression values between approximately 50 and 200, with fewer patients experiencing very high progression values. This indicates severe disease progression is relatively uncommon.

Histogram of bmi



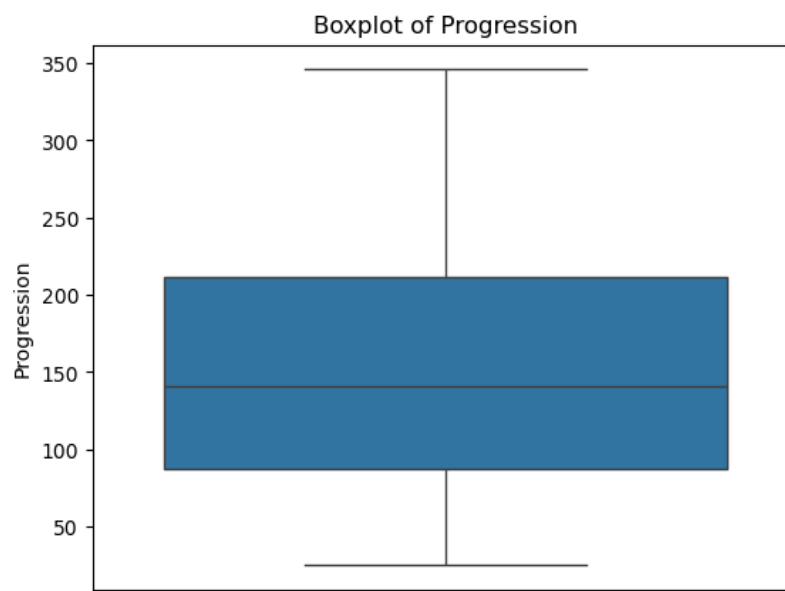
The BMI values appear to be centred close to zero, which is expected as the data is standardised. Most observations fall within a relatively narrow range, with only a small number of patients having particularly high or low BMI. There is no evidence of severe skewness or unusual clustering.

Boxplot of bmi



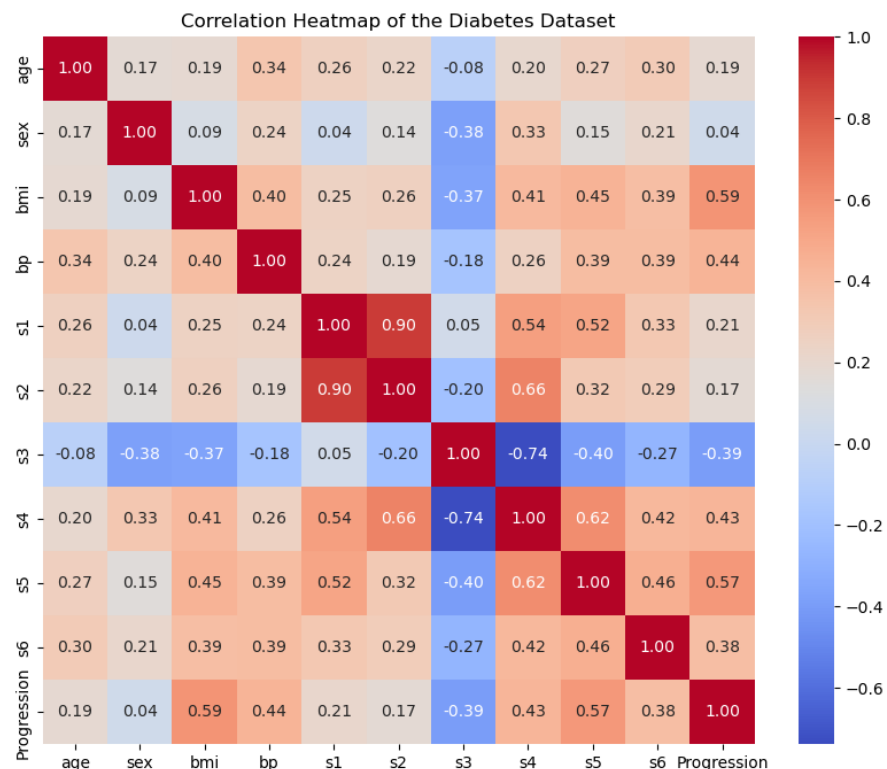
The boxplot shows the median BMI value, the interquartile range, and the overall spread of the data. The relatively short box indicates that the middle 50% of the observations are fairly concentrated. There are a small number of high outliers above the upper whisker, indicating a few patients have substantially higher BMI than the rest of the dataset.

Boxplot of Progression



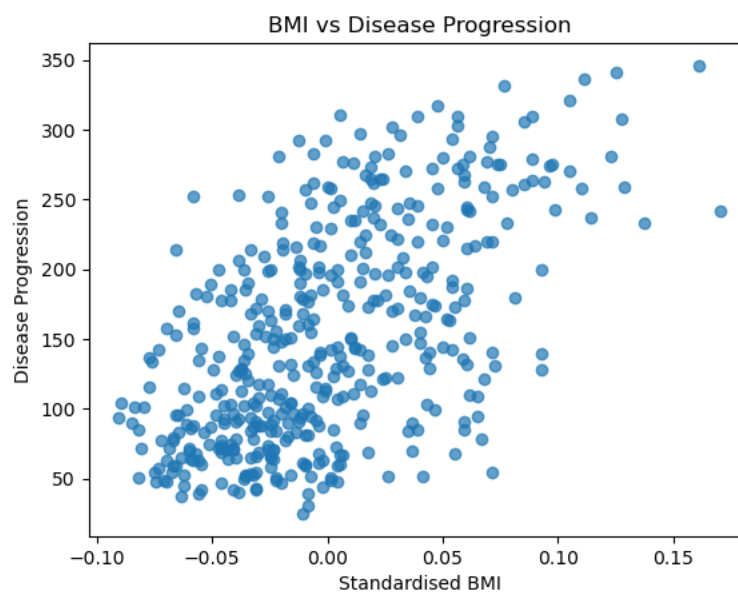
The progression values have a much larger spread than BMI, indicating greater variation between patients. The median lies close to the centre of the distribution, and there are no clear outliers beyond the whiskers.

Correlation heatmap



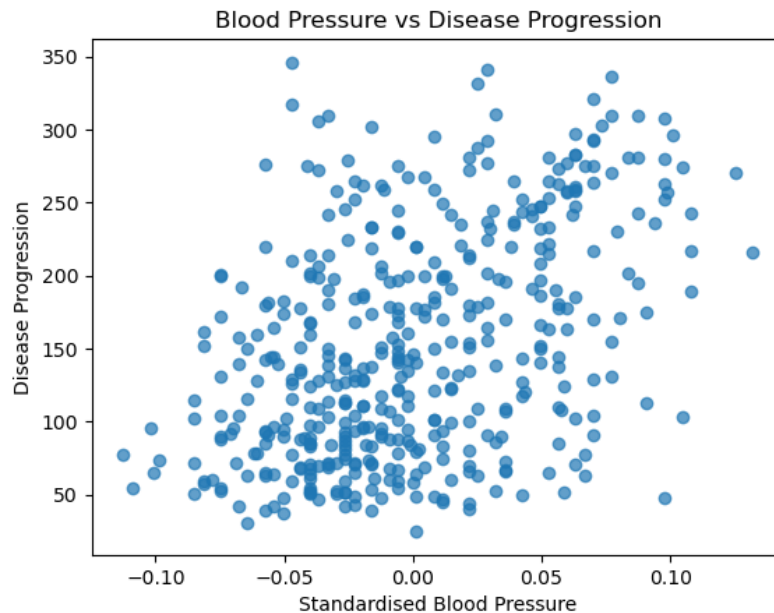
The heatmap shows that BMI and s5 have the strongest positive correlations with disease progression, suggesting that patients with higher BMI or triglyceride levels tend to experience greater disease progression. In contrast, HDL (s3) is negatively correlated with disease progression, indicating an inverse relationship. Most other variables exhibit weak to moderate correlations.

Scatter plot 1: bmi vs Progression



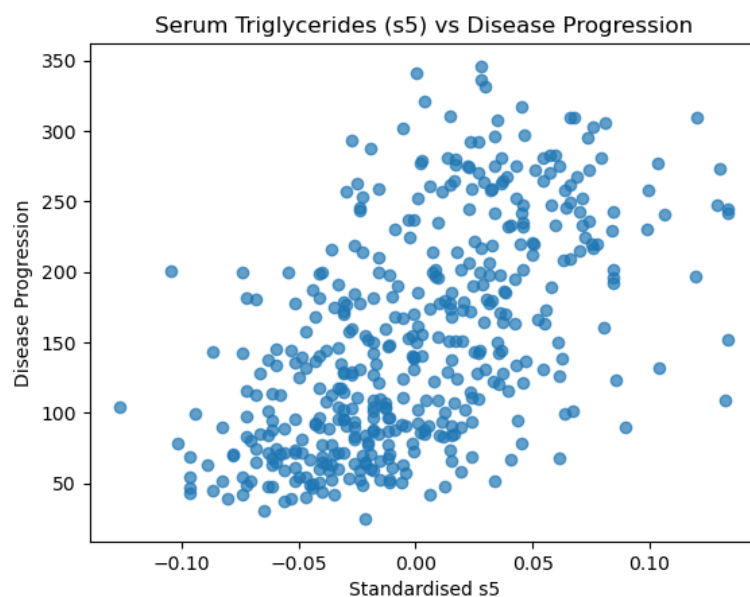
The scatter plot indicates a moderate positive relationship between BMI and disease progression. As BMI increases, progression tends to increase, although there is considerable variation. This suggests that BMI may be an important predictor but is not the sole factor influencing disease progression.

Scatter plot 2: bp vs Progression



The relationship between blood pressure and disease progression appears weaker than that of BMI. Although there is a positive trend, the points are widely scattered, indicating that blood pressure alone is not a strong predictor of progression.

Scatter plot 3: s5 vs Progression



The scatter plot shows a noticeable positive relationship between s5 (serum triglycerides) and disease progression. Higher triglyceride measurements are generally associated with higher progression values, although there is still considerable variability. This suggests that s5 may be one of the more informative predictors in the dataset.

Conclusion

Overall, the visualisations suggest that disease progression is most strongly associated with BMI and serum triglyceride levels (s5), both of which show moderate positive relationships with the target variable. Blood pressure also shows a positive association, although it is noticeably weaker. The distributions of the variables indicate that the dataset contains relatively few extreme values, and the boxplots reveal only a small number of BMI outliers. These findings suggest that BMI and s5 are likely to be the most informative predictors of disease progression within this dataset.

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